Enclosure 1: MTN Performance Work Statement (PWS)

1.0 Purpose

As noted in Instructions, Conditions, and Notices to Offerors, the Offeror shall use this Draft PWS as a basis, along with the Statement of Objectives (SOO) in Enclosure 1 and other references in the Request for Proposal (RFP), to develop and populate the Contract PWS to be submitted as Attachment A, MTN Performance Work Statement.

The Offeror must include the mandatory work statements prepopulated within this Draft PWS into their Contract PWS, either verbatim or using language that "meets the intent of." Where "meets the intent of" is used, the Offeror shall highlight the changes. The Offeror is advised that these mandatory statements are provided at a minimal level and require additional detail to describe the proposed effort.

Upon contract award, the selected Offeror's proposed PWS will be incorporated into the contract as Attachment A, MTN Performance Work Statement (PWS).

All portions of this Draft PWS that are in italicized font and underlined indicate instruction or additional information and should not be included in the Attachment A Contract MTN PWS.

2.0 General

This Performance Work Statement (PWS) defines the work to be performed by the Contractor for the Mars Telecommunications Network (MTN) project. The MTN project encompasses the design, development, integration, testing, delivery, launch operations support, and operational commissioning of a comprehensive Mars telecommunications and navigation relay system.

DRD: is Data Requirement Deliverable. The specific technical DRDs referenced herein are detailed in Attachment F, Data Requirement Deliverables.

- ***DRD-PM** series (Project Management)*
- ***DRD-MD** series (Mission Design)*
- ***DRD-SE** series (Systems Engineering)*
- ***DRD-SW** series (Software)*
- ***DRD-SMA** series (Safety & Mission Assurance)*
- ***DRD-COM** series (Communications)*

***DRD-FO** series (Flight Operations)*

The specific non-technical DRDs are detailed in Attachment H, CDRLS – Non-Technical Data Requirement Deliverables (DRDs). The Contractor shall deliver all DRDs specified in Attachment H, CDRLS Non-technical DRDs.

WBS: this Draft PWS employs a standard level 2 Work Breakdown Structure (WBS) used for space flight projects, but the Offeror may structure their proposed PWS at its discretion.

Standard Level 2 Work Breakdown Structure (WBS) Elements for Space Flight Projects

- *01 Project Management*
- *02 Systems Engineering*
- *03 Safety & Mission Assurance*
- *04 Science/Technology*
- *05 Payloads*
- *06 Spacecraft*
- *07 Mission Operations*
- *08 Launch Vehicle Services*
- *09 Ground Systems*
- *10 Systems Integration & Testing*

3.0 MTN Performance Work Statement

The Offeror shall fill in proposed work statements for this section that reflect their approach to meeting the MTN Core Contract Objectives (CCO) defined in Exhibit 1, Statement of Objectives (SOO). The Contractor shall utilize the mandatory statements below, either verbatim or language that "meets the intent of," in the Contract PWS.

1. Project Management

1.1. The Contractor shall deliver and implement all Project Management (**PM**) series deliverables.

1.1.1. **DRD-PM-01, Program Management Plan**.

1.1.2. **DRD-PM-02, Integrated Master Schedule (IMS)**.

1.1.3. **DRD-PM-03, Management Review Packages.**

1.1.4. **DRD-PM-04, Configuration and Data Management Plan.**

1.1.5. **DRD-PM-05, Risk Management Plan and Risk Management Reports.**

1.1.6. **DRD-PM-06, Major Review Plans and Documentation.**

1.1.7. **DRD-PM-07, Offeror's IT Security Management Plan.**

1.1.8. **DRD-PM-08, Insight and Oversight Implementation Plan.**

1.2. The Contractor shall ensure compliance with IT policy and regulations defined in the OCIO Requirements Coordination and Approval (ORCA): ORCA.GSFC.FY26.Q2.0033

1.2.1. IT Purchase Authorization

1.2.1.1. *The contractor shall ensure all IT products and services direct charged to the contract are approved through the OCIO's Commercial IT Request (CITR) Application. This includes, but is not limited to, FITARA, Supply Chain Risk Management (SCRM), IPv6, and 508 compliance. No purchases of commercial IT products or services should be made prior to coordination and approval by the OCIO. The Contractor shall report purchases monthly.*

1.2.1.2. *This contract shall only be used to purchase software, software maintenance, hardware, or hardware maintenance necessary for the performance of this contract. NASA shall own the license(s) or subscriptions for any software or applications purchased. All hardware shall be titled to NASA.*

1.2.2. Section 508 Conformance

1.2.2.1. *The contractor shall ensure that all products, platforms, services, and communications delivered as part of this work statement that are Information and Communication Technology (ICT) or contain ICT, conform to the Revised Section 508 of the Rehabilitation Act Standards, 36 C.F.R. § 1194.1 & Apps. A, C & D. The Standards may be found at https://www.access-board.gov/ict/.*

1.2.3. Records Management

1.2.3.1. *The Contractor shall maintain all records created for Government use or created in the course of performing the contract and/or delivered to, or under the legal control of the Government and must be managed in accordance with Federal law, including but not limited to, the Federal Records Act (44 U.S.C. chs. 21, 29, 31, 33), NARA regulations at 36 CFR Chapter XII Subchapter B, as well as NASA records policies (NPD 1440.6, NASA Records Management Program, and NPR 1441.1, NASA Records Management Program Requirements).*

1.2.3.2. *The Contractor shall ensure that NASA-owned/Contractor-held records are segregated from contractor-owned records and from non-record materials and report holdings of NASA records.*

1.2.3.3. *The Contractor shall immediately notify the Contracting Officer and NASA Records Officer upon discovery of any inadvertent or unauthorized removal, defacing, alteration, or destruction of Federal records. Destruction of Federal Records is EXPRESSLY PROHIBITED unless in accordance with records retention schedules or as directed by the Contracting Officer.*

1.2.3.4. *All Contractor employees assigned to this contract who create, work with, or otherwise handle Federal Records are required to complete mandatory NASA- provided records management training. The Contractor shall ensure that training has been completed according to agency policies, including initial training and any required annual or refresher training.*

1.2.3.5. *The Contractor shall incorporate the substance of this clause, its terms and requirements including this paragraph, in all subcontracts under this contract, and require written subcontractor acknowledgment of same.*

1.2.4. Use of Artificial Intelligence (AI) During Contract Performance

1.2.4.1. The contractor must disclose any potential or intended use of Artificial Intelligence (AI) technologies, tools, or systems in the performance of this contract. If it is unknown at the time of contract award whether AI will be used, the contractor must provide written notice to the Government prior to the deployment of any AI systems during contract performance. If AI technologies are introduced after the commencement of contract performance, the contractor must notify the contracting officer in writing prior to the release or deployment of any AI component, capability, or system in the NASA environment.

1.2.4.2. The Government reserves the right to review, approve, or disapprove the proposed use of AI technology in the performance of this contract. The contractor must obtain written approval from the contracting office before implementing AI systems.

1.2.4.3. If any subcontractor intends to use AI technologies in the performance of work under this contract, the contractor must ensure that similar notification and disclosure requirements apply to all subcontractors. The contractor is responsible for collecting the notification and disclosure requirements from subcontractors and submitting them to the Government.

2. *Systems Engineering*

2.1. The Contractor shall deliver and implement all Systems Engineering (**SE**) series deliverables.

- ***2.1.1.*** **DRD-SE-01, System Requirements.**
- ***2.1.2.*** **DRD-SE-02, System Specification.**
- ***2.1.3.*** **DRD-SE-03, Interface Control Documents and Drawings (ICDs).**
- ***2.1.4.*** **DRD-SE-04, Hardware Deliverable List.**
- ***2.1.5.*** **DRD-SE-05, Acceptance Data Package of Delivered Hardware.**
- ***2.1.6.*** **DRD-SE-06, Environmental Requirements Document.**
- ***2.1.7.*** **DRD-SE-07, Design Data Book.**
- ***2.1.8.*** **DRD-SE-08, Spacecraft Performance and System Resource Budget Reports.**
- ***2.1.9.*** **DRD-SE-09, Assembly, Integration and Test (AI&T) Plan.**
- ***2.1.10.*** **DRD-SE-10, Verification & Validation Plan.**
- ***2.1.11.*** **DRD-SE-11, Test Plans and Procedures.**
- ***2.1.12.*** **DRD-SE-12, Test Reports.**
- ***2.1.13.*** **DRD-SE-13, Verification Closure Notices and Validation Closure Notices/Requirements Verification Traceability Matrix.**
- ***2.1.14.*** **DRD-SE-14, Drawing and Document Trees.**
- ***2.1.15.*** **DRD-SE-15, Engineering Drawings, Models, and Associated Lists.**
- ***2.1.16.*** **DRD-SE-16, Design and Construction Standards Utilized.**
- ***2.1.17.*** **DRD-SE-17, Structural Analysis Reports.**
- ***2.1.18.*** **DRD-SE-18, Thermal Analysis Reports.**
- ***2.1.19.*** **DRD-SE-19, Power Analysis Report and Battery Usage Assessment.**
- ***2.1.20.*** **DRD-SE-20, Electromagnetic Interference (EMI) Electromagnetic Compatibility (EMC) Control Plan.**
- ***2.1.21.*** **DRD-SE-21, Worst Case Analysis.**

3. *Safety & Mission Assurance*

3.1. The Contractor shall provide NASA with on-request access to documents, records, and test data relevant to mission success and safety, including reliability analyses and heritage performance data.

3.2. The Contractor shall notify NASA within 24 hours of any major anomaly or test failure that has the potential to impact on-orbit performance, mission requirements, or the contracted schedule.

3.3. The Contractor shall notify NASA of any deviations from the Contractor's established internal standards and processes prior to implementation.

3.4. The Contractor shall notify NASA of changes to safety-critical or mission-critical hardware or software prior to implementation.

3.5. The Contractor shall provide NASA attendance at periodic program status reviews in an insight capacity.

3.6. The Contractor shall follow their existing workmanship processes.

3.7. The Contractor shall provide information of their existing processes to the Government for review.

3.8. The Contractor shall maintain a risk and hazard management process.

3.9. The Contractor shall provide NASA with periodic visibility into program risks and hazards that could impact on-orbit performance, mission success, safety, or the ability to meet the contracted delivery schedule, including mitigation plans and status.

3.10. The Contractor shall demonstrate compliance with applicable NASA and launch range safety requirements.

3.11. The Contractor shall identify and control prelaunch, launch, and ascent hazards associated with the flight system and its interfaces.

3.12. The Contractor shall maintain a supply chain risk management process.

3.13. The Contractor shall provide NASA with visibility into supply chain risks that could impact mission-critical or safety-critical components, subsystems, or software, including identification of sole-source dependencies, long-lead items, and counterfeit prevention measures.

3.14. The Contractor shall notify NASA of any supply chain disruption that could impact the contracted schedule or mission success.

3.15. The Contractor shall participate in the Government-Industry Data Exchange Program (GIDEP) and address GIDEP Alerts.

3.16. The Contractor shall address all NASA Advisories provided by the Government.

3.17. The Contractor shall have a documented process(es) for the establishment and operation of the Non-Conformance and Anomaly Review Board.

3.18. The Contractor shall invite the MTN Project Chief Safety and Mission Assurance Officer (CSO) or delegate to all major non-conformance meetings (i.e., Materials Review Boards (MRBs), Failure Review Boards (FRBs), or Anomaly Review Boards (ARBs)) as a voting member.

3.19. The Contractor shall identify major non-conformances as those that affect form, fit, function, require a software change, involve the presence of Foreign Object Debris (FOD), or those that the Contractor determines involve elevated technical or schedule risk.

3.20. The Contractor shall implement and document cybersecurity controls to protect the confidentiality, integrity, and availability of mission information and system resources.

3.21. The Contractor shall codify responses to the Exhibit 9 - Commercial Architecture Security Questionnaire (CASQ) in the PWS.

3.22. The Contractor shall deliver and implement all Safety & Mission Assurance (**SMA**) series deliverables.

3.22.1. **DRD-SMA-01, Request for Deviation or Waiver.**

3.22.2. **DRD-SMA-02, Safety and Mission Assurance (S&MA) Plan.**

3.22.3. **DRD-SMA-03, System Safety Analysis Report.**

3.22.4. **DRD-SMA-04, Failure Modes and Effects Analysis and Critical Items List / Single Point Failure.**

3.22.5. **DRD-SMA-05, EEE Parts Management Plan.**

3.22.6. **DRD-SMA-06, Parts Identification List.**

3.22.7. **DRD-SMA-07, Parts Stress/De-Rating Analysis.**

3.22.8. **DRD-SMA-08, Radiation Tolerance Report.**

3.22.9. **DRD-SMA-09, Nonconformance Processing and Corrective Action Plan and Nonconformance Reports.**

3.22.10. **DRD-SMA-10, Materials Identification and Usage List/Material Usage Agreement.**

3.22.11. **DRD-SMA-11, Planetary Protection Plan.**

3.22.12. **DRD-SMA-12, Safety Requirements Compliance Checklist (SRCC).**

4. Science/Technology

4.1. The Contractor shall document any technology development activities related to MTN, inclusive of technology demonstrations monthly.

5. Payloads

5.1. The Contractor shall describe (inclusive of mechanical, thermal, electrical, and command and data handling interfaces) all payload interfaces available for hosting NASA science payloads.

5.2. The Contractor shall define all constraints applied to hosted payloads.

5.3. The Contractor shall design the flight system to accommodate a NASA science payload that complies with MTN-REQ-049 in MTN-402-REQ-0001.

5.4. The Contractor shall integrate and test the flight system with the NASA science payload or a representative mass simulator.

5.5. The Contractor shall provide a payload command interface and a means for collecting, storing, and downlinking payload telemetry.

5.6. The Contractor shall separate payload telemetry (engineering and science) on the ground and deliver to the payload operations center in a timely fashion.

5.7. The Contractor shall implement fault detection and correction capability on the flight system that responds to anomalous thermal or electrical telemetry.

5.8. The Contractor shall deliver a low-fidelity spacecraft simulator to the payload team for interface testing.

6. Spacecraft

6.1. The Contractor shall deliver and implement all Communications (**COM**) series deliverables.

6.1.1. **DRD-COM-01, Communication System Analysis Report (DWE).**

6.1.2. **DRD-COM-02, Communication System Analysis Report (UHF Prox).**

6.1.3. **DRD-COM-03, Communication System Analysis Report (Advanced Prox).**

6.1.4. **DRD-COM-04, Spectrum Management and Compliance Information.**

7. Mission Operations

7.1. The Contractor shall describe their approach to extensibility and long-term system viability beyond 2035.

7.2. The Contractor shall deliver and implement all Mission Design (**MD**) series deliverables.

7.2.1. **DRD-MD-01, Mission Operations Plan/ConOps.**

7.2.2. **DRD-MD-02, Mission Design Document.**

7.3. The Contractor shall deliver and implement all Flight Operations (**FO**) series deliverables.

7.3.1. **DRD-FO-01, Operations Data Book/User Guide.**

7.3.2. **DRD-FO-02, Flight Operations Procedures and Procedures Data.**

7.3.3. **DRD-FO-03, Mission Operations Plans**

8. Launch Vehicle Services

8.1. The Contractor shall specify all relevant launch vehicle performance capabilities including payload mass capacity, trajectory performance, orbital insertion accuracy, and interface capabilities.

8.2. The Contractor shall identify and justify the selected launch vehicle and launch service provider based on mission requirements, schedule constraints, and cost considerations.

8.3. The Contractor shall define launch vehicle performance margins and demonstrate compliance with MTN mission requirements.

8.4. Launch Vehicle Integration

8.4.1. The Contractor shall specify launch vehicle interface and integration requirements including mechanical, electrical, thermal, and functional interfaces between the MTN spacecraft and launch vehicle.

8.4.2. The Contractor shall develop and maintain Interface Control Documents (ICDs) with the launch vehicle provider that define all physical, functional, and procedural interfaces.

8.4.3. The Contractor shall coordinate with the launch service provider to ensure compatibility of spacecraft design with launch vehicle capabilities and constraints.

8.4.4. The Contractor shall support payload integration activities at the launch site including spacecraft-to-launch vehicle adapter design, verification, and integration procedures.

8.5. Transportation and Launch Site Operations

8.5.1. The Contractor shall plan and execute safe transportation of the MTN spacecraft from the Contractor's facility to the launch site, including packaging, handling, environmental monitoring, and security measures.

8.5.2. The Contractor shall develop and implement transportation plans that address spacecraft protection, contamination control, environmental requirements, and security during transit to the launch site.

8.5.3. The Contractor shall coordinate launch site operations including spacecraft processing, integration with the launch vehicle, propellant loading, pre-launch checkout activities, and final closeouts & photography.

8.5.4. The Contractor shall provide personnel and support equipment necessary to perform spacecraft operations at the launch site including receiving, unpacking, inspection, processing, integration, and launch countdown support.

8.6. Launch Readiness and Mission Assurance

8.6.1. The Contractor shall support launch readiness reviews and certification activities.

8.6.2. The Contractor shall demonstrate spacecraft readiness for launch through completion of all verification activities, resolution of anomalies, and satisfaction of launch readiness criteria.

8.6.3. The Contractor shall identify and coordinate launch commit criteria, launch constraints, and abort procedures with NASA and the launch service provider.

8.6.4. The Contractor shall support launch countdown operations including spacecraft final configuration, monitoring, and go/no-go decision support.

8.7. Launch Campaign Coordination

8.7.1. The Contractor shall develop and maintain an integrated launch campaign schedule that coordinates spacecraft delivery, processing, integration, and launch activities with NASA and the launch service provider.

8.7.2. The Contractor shall coordinate launch manifest requirements, launch window constraints, and schedule dependencies with NASA and the launch service provider.

8.7.3. The Contractor shall support launch vehicle and range safety requirements including hazard analyses, safety documentation, and compliance with launch site safety regulations.

8.8. Environmental and Testing Support

8.8.1. The Contractor shall verify spacecraft compatibility with launch vehicle environments including vibration, acoustic, shock, thermal, and electromagnetic environments through analysis and testing.

8.8.2. The Contractor shall support combined systems testing and integrated testing activities with the launch vehicle as required by the launch service provider.

8.9. Contingency and Anomaly Support

8.9.1. The Contractor shall develop contingency plans for launch delays, anomalies, or scrubs including spacecraft safing, de-integration, and re-integration procedures.

8.9.2. The Contractor shall provide engineering support to investigate and resolve any spacecraft or interface anomalies identified during launch site operations.

9. Ground Systems

9.1. The Contractor shall implement software assurance practices consistent with their internal standards or an equivalent industry standard.

9.2. The Contractor shall identify and classify all safety-critical and mission-critical flight software.

9.3. The Contractor shall disclose to NASA any known software defects, limitations, or anomaly history that could affect mission success, safety, or schedule.

9.4. The Contractor shall provide NASA with visibility into software problem/defect tracking for flight software upon request.

9.5. The Contractor shall deliver and implement all Software (**SW**) series deliverables.

9.5.1. **DRD-SW-01, Software Requirements Specification (SRS).**

9.5.2. **DRD-SW-02, Master Data Dictionary.**

9.5.3. **DRD-SW-03, Software Management and Development Plan.**

9.5.4. **DRD-SW-04, Software Design Description/Version Description Document.**

10. Systems Integration & Testing

10.1. The Contractor shall develop and implement comprehensive verification plans that demonstrate how all requirements specified in Attachment B, Mars Telecommunications Network (MTN) Objectives and Requirements, and all derived lower-level requirements will be verified throughout the MTN development lifecycle.

10.2. The Contractor shall define the verification methods (analysis, inspection, demonstration, and test) for each requirement.

10.3. The Contractor shall establish the verification sequence, environments, success criteria, and responsible organizations for all verification activities.

10.4. The Contractor shall address the core philosophy of the verification approach and trades that are in play to balance the tension between a rigorous verification program and the aggressive delivery schedule.

10.5. The Contractor shall identify a process by which deviations and waivers to the verification plan will be approved by NASA.

10.6. The Contractor shall address all phases of the program including design verification, qualification testing, acceptance testing, integration and test, launch readiness verification, and on-orbit checkout and commissioning in the verification plans.

10.7. The Contractor shall demonstrate how the verification approach will confirm that the MTN design and construction activities achieve the required performance, reliability, safety, and mission assurance objectives.

10.8. The Contractor shall provide NASA with visibility into verification methods, criteria, and results through the verification plan.

10.9. The Contractor shall include appropriate insight into verification activities to support the certification process and ensure that the MTN meets all performance objectives prior to operational deployment.

10.10. The Contractor shall provide access to key verification events and test activities as appropriate to enable Government oversight and risk assessment.